

Bridging the Heisenberg Spin Model to Spatial IETS dI/dV Maps Triangulene–Phenalenyl Nanographene Polymer

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1 Introduction: from a discrete spin lattice to a spatial STM observable

The low-energy magnetism of the phenalenyl–triangulene copolymer $P-T-P-[P-T-P]_{N-1}$ is captured by a Heisenberg Hamiltonian on a chain of $4N$ spin- $\frac{1}{2}$ sites. With the ordering $[P, T_a, T_b, P]$ inside each monomer,

$$H = \sum_{m=0}^{N-1} \left[J_{TP} (\mathbf{S}_{4m} \cdot \mathbf{S}_{4m+1} + \mathbf{S}_{4m+2} \cdot \mathbf{S}_{4m+3}) + J_T \mathbf{S}_{4m+1} \cdot \mathbf{S}_{4m+2} \right] + J_{PP} \sum_{m=0}^{N-2} \mathbf{S}_{4m+3} \cdot \mathbf{S}_{4m+4}, \quad (1)$$

with $J_{PP} = 38$ meV, $J_{TP} = 40$ meV and $J_T = -500$ meV. Exact diagonalization yields a total-singlet ground state $|0\rangle$ ($S = 0$) and, as the first magnetic excitation, a triplet multiplet $\{|1_m\rangle\}_{m=-1,0,+1}$ ($S = 1$) separated by the spin gap $\Delta = E_1 - E_0$.

Inelastic electron tunneling spectroscopy (IETS) in a low-temperature STM probes exactly this excitation: when the bias energy $e|V|$ exceeds Δ , a tunneling electron can flip the local spin and open an inelastic channel, producing a symmetric step in dI/dV at $e|V| = \Delta$. Crucially, the *intensity* of that step is spatially resolved, because it is governed by the local spin matrix element between $|0\rangle$ and $|1_m\rangle$ at the position of the tip. The purpose of this note is to make explicit the bridge between the discrete spin model (1) and the continuous $dI/dV(x, y)$ image recorded in experiment.

2 Transition weights from the spin operators

To second order in the tip–sample coupling, the inelastic spin-flip conductance at site i is proportional to the Fermi-golden-rule rate summed over the degenerate final states of the excitation. For a spin- $\frac{1}{2}$ scatterer the relevant tunneling vertex is the local spin operator $\mathbf{S}_i = (S_i^x, S_i^y, S_i^z)$, so the *per-site transition weight* reads

$$W_i = \sum_m \sum_{\alpha=x,y,z} |\langle 1_m | S_i^\alpha | 0 \rangle|^2, \quad (2)$$

where the inner sum runs over the three Cartesian spin components and the outer sum over the members m of the final triplet multiplet. Writing out the three terms for a single final state,

$$W_i = |\langle 1 | S_i^x | 0 \rangle|^2 + |\langle 1 | S_i^y | 0 \rangle|^2 + |\langle 1 | S_i^z | 0 \rangle|^2, \quad (3)$$

which is the form used operationally. Summing over the full multiplet in (2) is important: because the triplet is three-fold degenerate, any individual $|1_m\rangle$ returned by the diagonalizer is defined only up to an arbitrary unitary rotation within the multiplet, whereas the sum in

Eq. (2) is invariant. Indeed, since $|0\rangle$ is a spin singlet ($S = 0$, $S_z = 0$), the selection rules make the assignment transparent: S_i^z connects $|0\rangle$ only to the $S_z = 0$ triplet member, while S_i^x, S_i^y (equivalently $S_i^\pm = S_i^x \pm iS_i^y$) connect it to the $S_z = \pm 1$ members. The multiplet sum therefore collects the complete spin-flip spectral weight localized on site i ,

$$W_i = \sum_m \langle 0 | \mathbf{S}_i | 1_m \rangle \cdot \langle 1_m | \mathbf{S}_i | 0 \rangle, \quad (4)$$

a real, rotationally invariant quantity that plays the role of a local “inelastic magnetic susceptibility” of the transition. For the $N = 3$ chain the W_i are largest on the triangulene sites T_a, T_b (which carry the dominant $S = 1$ character) and are symmetric under the mirror operation of the chain, consistent with the symmetry of Eq. (1).

3 Phenomenological spatial mapping (Tersoff–Hamann)

To turn the site-resolved weights W_i into a real-space image we adopt the Tersoff–Hamann description of STM, in which the tunneling current at tip position $\mathbf{r}_{\text{tip}}$ is proportional to the local density of states of the sample evaluated at the tip. For an s -wave tip the sample orbitals decay exponentially into the vacuum barrier,

$$\psi(\mathbf{r}) \sim \exp(-\kappa |\mathbf{r} - \mathbf{r}_\alpha|), \quad \kappa = \frac{\sqrt{2m_e\phi}}{\hbar}, \quad (5)$$

where ϕ is the effective work function and κ the inverse vacuum decay length (of order $1 \text{ \AA}^{-1}$). Each magnetic site i is spatially distributed over the carbon atoms α that build the corresponding π -radical (one phenalenyl, or one half of a triangulene). Placing the tip at constant height z_0 above the molecular plane, $\mathbf{r}_{\text{tip}} = (x, y, z_0)$ and $\mathbf{r}_\alpha = (x_\alpha, y_\alpha, 0)$, and weighting each atomic contribution by the transition weight of its site, the simulated inelastic conductance map is

$$\boxed{\frac{dI}{dV}(x, y) \propto \sum_{i=0}^{4N-1} W_i \sum_{\alpha \in \text{site } i} \exp\left(-\kappa \sqrt{(x - x_\alpha)^2 + (y - y_\alpha)^2 + z_0^2}\right)}. \quad (6)$$

Equation (6) is the central phenomenological object: the spin physics enters entirely through the weights W_i of Eq. (2), while the exponential kernel of Eq. (5) supplies the spatial broadening and the constant-height blurring set by z_0 . Larger κ or smaller z_0 sharpen the image toward the atomic positions; smaller κ or larger z_0 produce the smooth lobes characteristic of experimental constant-height maps.

Bias condition. The map (6) is the spatial fingerprint of the *single* excitation of energy Δ . It corresponds to tuning the STM bias to the inelastic threshold,

$$eV = \Delta = E_1 - E_0, \quad (7)$$

i.e. the voltage at which the singlet→triplet spin-flip channel opens. Recording dI/dV at this bias and rastering the tip over the molecule yields, within this model, the image predicted by Eq. (6), directly comparable to LT-STM IETS data.